▶ DAVID PYM, *Generalized Proof-theoretic Validity.*
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In proof-theoretic semantics (P-tS) [1, 2, 3, 4], the validity of formulae and of proofs is judged not through interpretation in models, but rather through inference in systems of pre-logical base rules. Within P-tS, it is often convenient to distinguish its theory of the validity of proofs, henceforth called proof-theoretic validity (P-tV), and its theory of the validity of formulæ, henceforth called base-extension semantics (B-eS). Both P-tV and B-eS employ the idea of a pre-logical *base* (a set) of inference rules. In the simplest set-up, base rules refer only to atomic propositional formulæ p (and sets P thereof) and take forms (in the style of 'natural deduction') such as 'production' rules of the form $p_1, \ldots, p_k \to p$, or their generalization to include dischargeable hypotheses, and more — see, for example, Sandqvist's discussion in [5] — with the choice of the form or rules having a profound effect on the logical strength of the semantics.

Prawitz's theory of P-tV [1, 2, 4] judges the validity of proofs relative to a base $\mathcal{B}$ and a justification $\mathbf{J}$, which maps proof-structures (i.e., proof-like structures) to proof-structures. $\langle \mathbf{J}, \mathcal{B} \rangle$-validity is now defined relative to a class of canonical proofs $\mathbf{C}$ as follows: a proof-structure Φ is $\langle \mathbf{J}, \mathcal{B} \rangle$-valid — that is, represents a proof — if either $\Phi \in \mathbf{C}$ or if $\mathbf{J}$ can be applied to Φ to yield an element of $\mathbf{C}$. Atoms are proved in the base. This definition gives rise to a consequence relation [4]. In the setting of intuitionistic logic, the required canonical form for proofs is essentially normal form in the sense of the theory of natural deduction and the closely associated BHK semantics. These issues are discussed, for example, by Schroeder-Heister [4].

I argue that the ideas of P-tV arise more naturally given three inter-related changes of perspective. First, the underlying logical model based on *deductive* reasoning should be replaced with one based on *reductive* reasoning (e.g., [6] and references therein) — in which rules of inference are read from putative conclusion to sufficient premisses. Second, the primary objects of proof should be *sequents* (cf. [4]). Third, the notion of canonical form should *not* be constrained to resemble the normal forms of natural deduction. I will argue that such a theory would be quite natural for practical reasoning (e.g., [7]) and have useful connections with B-eS (see, e.g., [8]).

[1] Dag Prawitz, *Ideas and Results in Proof Theory*, ***Studies in Logic and the Foundations of Mathematics***, 63, 235–307.

[2] Dag Prawitz, *Towards a Foundation of a General Proof Theory*, ***Studies in Logic and the Foundations of Mathematics***, 74, 225–250.

[3] Michael Dummett, ***The Logical Basis of Metaphysics***, Oxford University Press, 1991.

[4] Peter Schroeder-Heister, M. Pelis (editor) ***The Logica Yearbook 2007***, *Proof-theoretic Versus Model-theoretic Consequence*, Filosofia, 2008.

[5] Tor Sandqvist, *Atomic Bases and the Validity of Peirce's Law*, ***The meaning of proofs: Celebrating World Logic Day, E. Pimentel and D. Pym (editors)*** , `https://sites.google.com/view/wdl-ucl2022/home`, 2022

[6] Alexander Gheorghiu and David Pym, *Semantic Foundations of Reductive Reasoning*, ***Topoi***, `https://doi.org/10.1007/s11245-025-10211-6`, 2025.

[7] Alexander Gheorghiu and David Pym, *Definite formulae, negation-as-failure, and the base-extension semantics for intuitionistic propositional logic*, ***Bulletin of the Section of Logic***, `https://doi.org/10.18778/0138-0680.2023.16`, 2023.

[8] Alexander Gheorghiu and David Pym, *From Proof-theoretic Validity to Base-extension Semantics for Intuitionistic Propositional Logic*, ***Studia Logica***, `https://doi.org/10.1007/s11225-024-10163-9`, 2025.